Sets & Venn Diagrams

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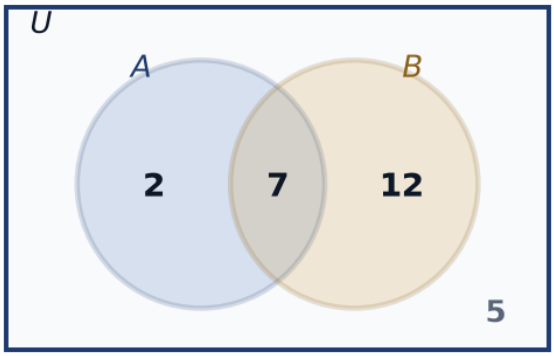
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the diagram, ABC is a triangle with angle ABC = 90 degrees. AD is the altitude to the hypotenuse BC. Given that BD = 4 cm and DC = 9 cm, find the length of the altitude AD.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows triangle ABC and its image A'B'C' after a single transformation. Determine the coordinates of A' if A is at $(1, 2)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The points A = (2, k), B = (8, 4) and C = (5, 7) form a right-angled triangle with the right angle at B. Find the value of k.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle ABC, $AB = 8$ cm, $BC = 11$ cm, and angle $BAC = 45^\circ$. Find the obtuse angle $\angle ACB$ to 1 decimal place.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a cyclic quadrilateral $ABCD$, AB is extended to point E. If angle $\angle CBE = 74^\circ$, what is the size of interior angle $\angle ADC$?

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle PQR, angle $P = 40^\circ$, angle $Q = 75^\circ$, and side $q = 12$ cm. Use the sine rule to find the length of side p.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle ABC, $a = 5$ cm, $b = 7$ cm, and $c = 8$ cm. Calculate the size of angle B to the nearest degree.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The diagram shows points A, B, and C. Point A' is the image of point A after a translation. Find the coordinates of A'.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The line segment AB has a midpoint M at $(4, 7)$. Point A is located at $(1, 3)$. Find the coordinates of point B, denoted as (x, y).

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The coordinates of point A are $(-2, 3)$ and the coordinates of point B are $(4, -1)$. Point M is the midpoint of the line segment AB. Find the coordinates of M.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A solid metal cone with radius 6 cm and height 12 cm is melted down and recast into solid spheres of radius 2 cm. How many complete spheres can be made?

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A bag contains 5 red balls and 3 blue balls. A ball is chosen at random from the bag. What is the probability that the ball chosen is red?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the diagram below, lines AB and CD are parallel. A transversal line intersects them at points E and F respectively. If the angle $\text{AEF} = 50^\circ$, find the measure of angle EFC.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cuboid has length 10 cm, width 6 cm, and height 4 cm. Calculate the angle that the space diagonal makes with the base of the cuboid.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Transformations & Vectors

A bag contains 6 white balls and 4 black balls. Two balls are drawn at random without replacement. What is the probability of drawing one ball of each colour?

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

On a histogram, the class $0 \leq w < 10$ has a height of 1.4 cm and represents a frequency of 28. What frequency is represented by a bar for class $10 \leq w < 25$ with a height of 2.0 cm?

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Functions

Ship X sails from port P on a bearing of 040° for 60 miles to point A. Ship Y sails from P on a bearing of 135° for 80 miles to point B. Calculate the distance between A and B to 1 decimal place.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number

A cuboid has a length of USD $8cm, awidthofUSD5$ cm, and a height of USD 3 cm. Calculate the volume of the cuboid.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

The coordinates of point P are $(2, 5)$ and the coordinates of point Q are $(6, 13)$. Calculate the gradient of the line segment PQ.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Coordinate Geometry

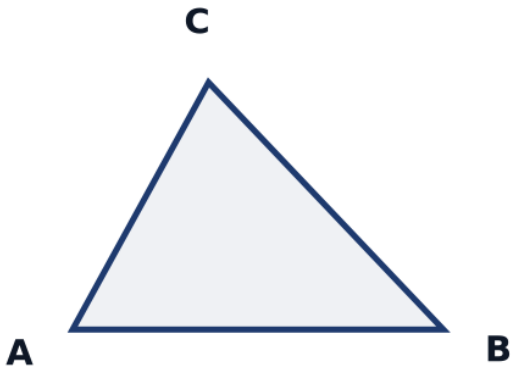
In the diagram, a transversal line intersects two parallel lines. Find the value of x.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying transformations and congruence, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 103°

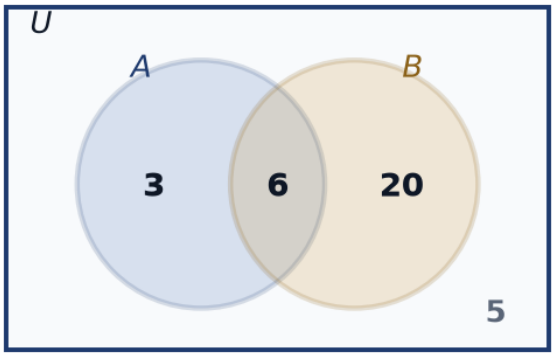
b. 113°

c. 95°

d. 148°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

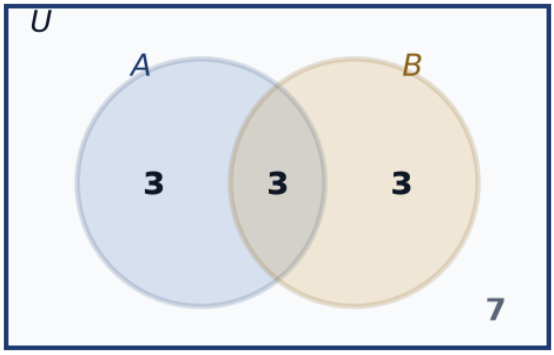
c. 24 cm

d. 54 cm

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?



a. 9

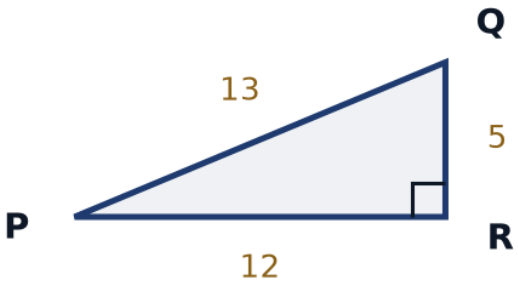
b. 14

c. 12

d. 15

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying congruent triangle tests (extended), $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 102°

b. 112°

c. 94°

d. 147°

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for scale factors consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 5$ in scale factors, evaluate $f(4)$.

a. 29

b. 35

c. 24

d. 33

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying similar triangle tests, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 93°

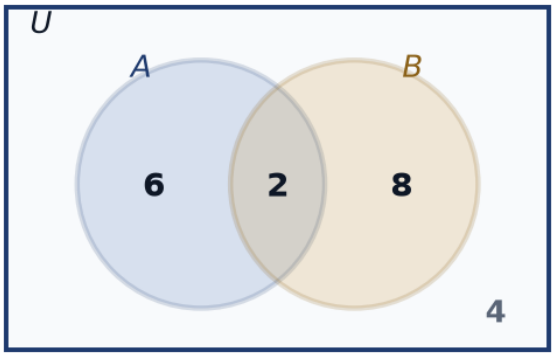
b. 103°

c. 85°

d. 145°

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Universal set $\mathscr{E} = \{x : x \text{ is an integer}, 1 \le x \le 10\}$. Set $A = \{2, 3, 5, 7\}$. List the elements of A'.



a. $\{1, 4, 6, 8, 9, 10\}$

b. $\{4, 6, 8, 9, 10\}$

c. $\{1, 4, 6, 8, 10\}$

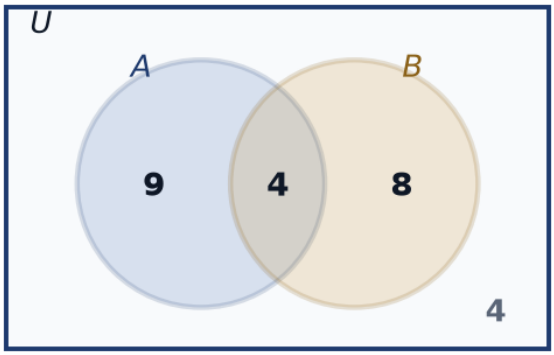
d. $\{2, 3, 5, 7\}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Universal set $\mathscr{E} = \{x : x \in \mathbb{Z}^+, 1 \le x \le 12\}$. Set $A = \{2, 3, 5, 7, 11\}$. How many elements are in A'?



a. 7

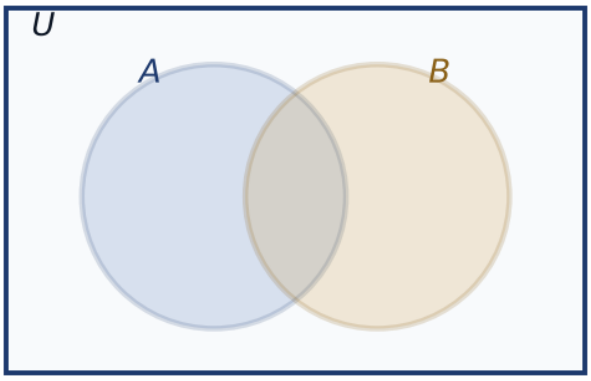
b. 6

c. 5

d. 12

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which set notation represents the region inside A and outside B?



a. $A \cap B'$

b. $A' \cap B$

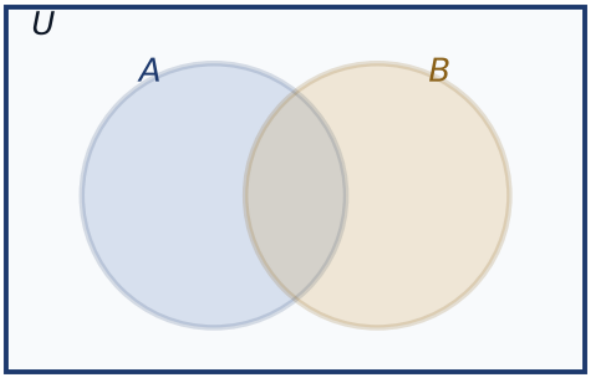
c. $A \cup B$

d. $(A \cup B)'$

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which set notation represents the region inside B and outside A?



a. $A' \cap B$

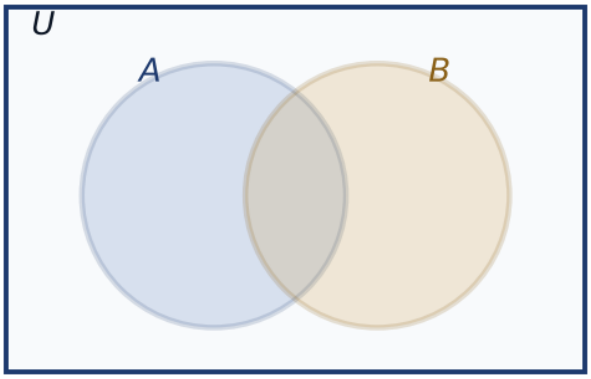
b. $A \cap B'$

c. $A \cap B$

d. $A' \cup B$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which set notation represents the region outside both A and B?



a. $(A \cup B)'$

b. $A \cap B$

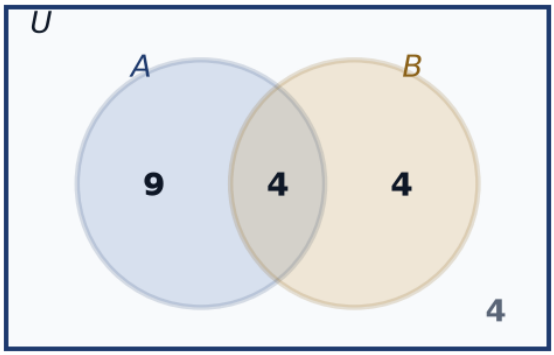
c. $A' \cap B$

d. $A' \cup B'$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If set $A = \{2, 4, 6\}$ and set $B = \{1, 3, 5\}$, what is $A \cap B$?



a. $\varnothing$

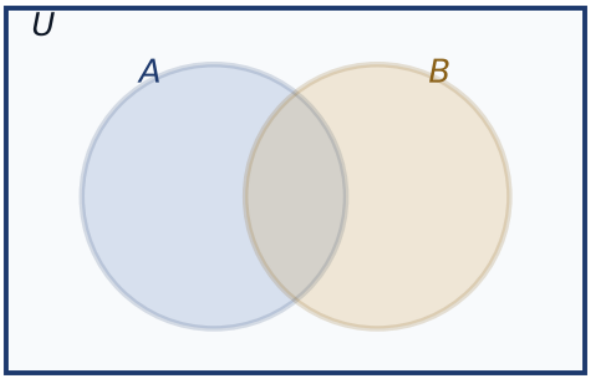
b. $\{0\}$

c. $\{1, 2, 3, 4, 5, 6\}$

d. $\{\varnothing\}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which set notation represents the region inside both A and B?



a. $A \cap B$

b. $A \cup B$

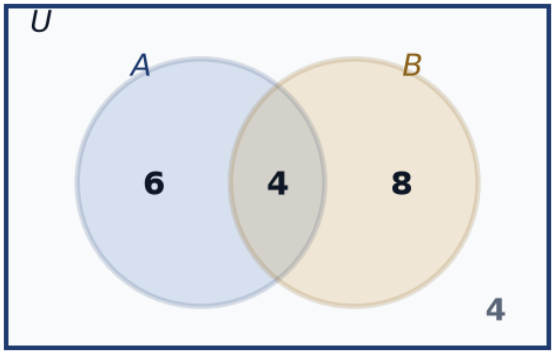
c. $A' \cap B'$

d. $A \setminus B$

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35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In a Venn diagram, region $A \cap B$ has 9 elements, $A \cap B'$ has 14, and $A' \cap B$ has 11. If $n(\mathscr{E}) = 40$, find $n((A \cup B)')$.



a. 6

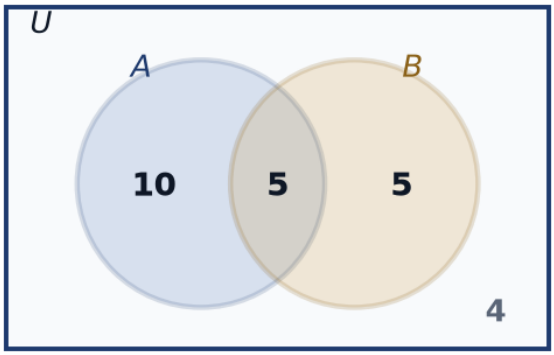
b. 4

c. 8

d. 10

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

If $A \subset B$, which of the following is always equal to $A \cap B$?



a. A

b. B

c. $\varnothing$

d. A'

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a survey of 50 tourists, 30 visited museum M, 25 visited palace P, and 20 visited art gallery G. 12 visited both M and P, 10 visited both P and G, 8 visited both M and G, and 5 visited all three. How many tourists visited NONE of the three attractions?

a. 0

b. 5

c. 8

d. 10

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number

Which of the following statements represents 'the empty set'?

a. $\varnothing$

b. $\{0\}$

c. $\mathscr{E}$

d. A'

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Algebra & Graphs

In a class of 30 students, 18 study French (F), 15 study German (G), and 7 study both French and German. How many students study neither French nor German?

a. 4

b. 7

c. 26

d. 11

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Coordinate Geometry

Set A has 15 elements and set B has 10 elements. What is the minimum possible number of elements in $A \cup B$?

a. 15

b. 25

c. 13

d. 5

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Mensuration

Which set notation represents the shaded region outside set A but inside set B?

a. $A' \cap B$

b. $A \cap B'$

c. $(A \cup B)'$

d. $A' \cup B$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Geometry

In a group of 100 people, 45 read newspaper A, 40 read B, 35 read C. 15 read A and B, 12 read B and C, 10 read A and C, and 5 read all three. How many read NONE?

a. 12

b. 17

c. 88

d. 9

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Trigonometry

In a class, 5 study only French, 7 study only Spanish, 5 study both, and 2 study neither. Find the total class size.

a. 23

b. 15

c. 17

d. 19

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Transformations & Vectors

In a survey of 40 students, 24 study French, 18 study German, and 10 study both. How many study neither?

a. 8

b. 12

c. 32

d. 6

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Probability & Statistics

Given $n(\mathscr{U}) = 50$, $n(A) = 25$, and $n(A \cup B) = 40$. If A and B are mutually disjoint ($A \cap B = \varnothing$), find $n(B)$.

a. 15

b. 25

c. 10

d. 20

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